

AYANDA THABANI MTHETHWA

Junior Software Developer · Full-Stack Web Development

ayoh45thabany@gmail.com | 063 439 9328 / 079 588 0596 | Ivory Park Ext 2, Midrand, 1632

PROFESSIONAL SUMMARY

Results-driven Junior Software Developer with proven, hands-on experience building and deploying production-ready full-stack web applications across multiple technology stacks. Holds a National Diploma in Software Development (TUT). Delivered a live citizen-facing service delivery portal (React / Node.js) and a production-deployed real-time chat platform (Angular 21 / ASP.NET Core 9 / SignalR) featuring WebRTC video calling, JWT authentication, and role-based access control. Adept at writing clean, scalable code and collaborating within Agile teams. Actively seeking a Junior Developer role to contribute meaningfully to innovative software solutions.

TECHNICAL SKILLS

Frontend	Angular 21, React, TypeScript, JavaScript (ES6+), HTML5, CSS3, Angular Signals
Backend	ASP.NET Core 9 (C#), Node.js, Python, RESTful APIs, SignalR (WebSockets)
Database	PostgreSQL, MySQL, Entity Framework Core 9
Auth & Security	JWT Bearer Tokens, BCrypt password hashing, Role-Based Access Control (RBAC)
Real-Time	SignalR, WebRTC (video calling), WebSocket communication
Tools	Git / GitHub, Axios, Agile / Scrum, Swagger / OpenAPI, Docker, Render
Soft Skills	Problem-solving, attention to detail, fast learner, team collaboration

WORK EXPERIENCE

Full-Stack Software Developer | ICEP — Tshwane University of Technology *Jul 2024 – Jun 2025*

- Developed and deployed a full-stack citizen service delivery portal used by community members, supervisors, managers, and administrators — demonstrating end-to-end project ownership.
- Built responsive, reusable React components that improved UI consistency and significantly reduced duplicate code across all user roles.
- Integrated the frontend with backend RESTful APIs using Axios, enabling reliable, efficient data communication between layers.
- Designed and implemented Role-Based Access Control (RBAC) for four distinct user types, enforcing data security and appropriate permission levels.
- Operated in an Agile-style environment with regular stakeholder reviews and milestone-driven delivery, meeting all project deadlines.

KEY PROJECTS

ChatSystem — Real-Time Messaging Platform *Angular 21 · ASP.NET Core 9 (C#) · SignalR · PostgreSQL · WebRTC · JWT · Docker · Render*

Live App: <https://chat-app-frontend-0ncz.onrender.com/>

- Built a production-deployed, full-stack real-time chat platform with persistent 1-on-1 messaging, WhatsApp-style read receipts (single/double blue ticks), and live typing indicators — all powered by SignalR WebSockets with no polling.
- Implemented 1-on-1 WebRTC video calling with incoming call modals, in-call overlay, and real-time WebRTC signaling routed through the PresenceHub.
- Secured the platform with JWT Bearer authentication, BCrypt password hashing, and RBAC — with Angular route guards on the client and [Authorize(Roles)] attributes enforcing permissions server-side.

- Built a dedicated Admin Dashboard enabling full user management, real-time online status monitoring, and access to any conversation thread — with self-delete protection for admin accounts.
- Engineered multi-tab presence tracking using a ConcurrentDictionary of connection IDs per user, ensuring accurate online/offline status across devices.
- Deployed the full stack to Render using a multi-stage Docker build, a managed PostgreSQL database, and a Render Blueprint (render.yaml) for infrastructure-as-code deployment.
- Migrated the database from MySQL to PostgreSQL and upgraded the stack from Angular 18 / .NET 8 to Angular 21 / .NET 9.

Extreme Service Delivery Portal *React · Node.js · JavaScript · MySQL*

- Citizen-facing platform enabling community members to report service delivery issues directly to government officials.
- Implemented RBAC for four user roles: citizen, supervisor, manager, and admin — demonstrating scalable, secure architecture design.
- Delivered a fully responsive UI with integrated backend APIs supporting real-time data management across all user types.

Web-Based Ticketing System *React · Node.js · JavaScript · MySQL · RESTful APIs*

- Built a complete internal service request management system with ticket submission, assignment, tracking, and resolution workflows.
- Designed the full database schema from scratch and integrated REST APIs for seamless, reliable data communication.
- Implemented RBAC covering the complete ticket lifecycle, ensuring appropriate access at each stage of the workflow.

Personal Portfolio Website *HTML5 · CSS3 · JavaScript*

- Designed and developed a personal portfolio website showcasing projects, skills, and contact information — demonstrating frontend proficiency and attention to UI/UX.

EDUCATION

National Diploma in IT: Software Development | Tshwane University of Technology (TUT) 2024

National Senior Certificate (Matric) | Nhlungwane High School 2012

CERTIFICATIONS

- National Diploma in Software Development — Tshwane University of Technology, 2024

PERSONAL DETAILS

Date of Birth: 10 March 1995

Nationality: South African

Gender: Male

Home Language: IsiZulu

REFERENCES

Gustavo Mamosadi

Supervisor, ICEP | Tshwane University of Technology

Mamosading@tut.ac.za | 072 282 6821

Mr V.N. Ranko

WIL Coordinator & Lecturer, Tshwane University of Technology

RankoVN@tut.ac.za | 012 382 9112